Supplementary Material

1 Search profile

Keyword search strategy used to find published nanotoxicological studies.

Field Delimiter	Search Field	Search Criterion	Search Term
	All Fields	Contains	nanotox*
OR	All Fields	Contains	fulleren* AND toxic*
OR	All Fields	Contains	carbo nanotube* AND toxic*
OR	All Fields	Contains	bucky ball* AND toxic
OR	All Fields	Contains	nanotube* AND toxic
OR	All Fields	Contains	nanoparticle* AND toxic*
OR	All Fields	Contains	nanomat* AND toxic*
OR	All Fields	Contains	nano* AND toxic
OR	Year	Contains	2013[1]

[1] for each year a separate search has been carried out. For yearly results see (1).
*: wildcard for the search words

Using this profile roughly 3'000 to 5'000 publications per year have been found. For the systematic review on amorphous silica all 18'162 records have been selected following the PRISMA flow diagram shown in Figure 1.

1. Krug HF. The uncertainty with nanosafety: Validity and reliability of published data. Colloids Surf B Biointerfaces. 2018;172:113-117.

2 Complete Reference List

Red: publications with quality score 0,8 (good reliability) and 1 (very good reliability)

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2.1 List of publications with silica-coated materials

(Excerpt from complete list above)

Chia, SL et al. (2016). **Reducing ZnO nanoparticles toxicity through silica coating.** *Heliyon*, 2(10): e00177. **https://doi.org/10.1016/j.heliyon.2016.e00177**

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Ha, SW et al. (2013). **Long-Term Monitoring of the Physicochemical Properties of Silica-Based Nanoparticles on the Rate of Endocytosis and Exocytosis and Consequences of Cell Division.** *Soft Mater*, 11(2): 195-203. **https://doi.org/10.1080/1539445X.2012.617641**

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3 Datapoints for figures are taken from the following studies

(Numbers from the complete list above; number in red: high-quality studies)

Figure 2A and 2B. Values from: 1, 2, 5, 7, 9, 12, 13, 14, 15, 18, 24, 27, 30, 31, 33, 38, 41, 42, 43, 45, 52, 53, 61, 63, 64, 65, 68, 70, 72, 73, 76, 79, 80, 82, 84, 85, 87, 88, 89, 90, 91, 94, 97, 100, 100, 101, 103, 108, 109, 112, 113, 116, 118, 121, 122, 123, 124, 126, 127, 129, 130, 131, 134, 136, 145, 145,

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Figure 3A and 3B. Values from: 2, 4, 8, 10, 13, 14, 19, 20, 24, 26, 30, 36, 48, 52, 53, 63, 69, 72, 73, 76, 80, 87, 90, 91, 100, 109, 113, 115, 116, 122, 123, 124, 127, 131, 134, 138, 144, 145, 148, 152, 153, 155, 156, 163, 168, 169, 176, 177, 182, 184, 186, 188, 189, 193, 196, 201, 206, 207, 211, 216, 228, 249, 256, 263, 276, 281, 282, 286, 295, 296, 297, 316.

Figure 4A and 4B. Values from: 2, 7, 8, 9, 10, 12, 24, 28, 36, 38, 41, 43, 52, 53, 62, 69, 70, 76, 85, 88, 90, 91, 92, 97, 109, 113, 115, 119, 123, 125, 126, 127, 145, 148, 149, 159, 163, 164, 168, 169, 171, 176, 184, 207, 211, 215, 216, 224, 228, 231, 248, 255, 261, 264, 268, 276, 281, 282, 294, 296, 298, 304, 307, 308, 316

Figure 5A and 5B. Values from: 1, 15, 19, 20, 24, 26, 28, 29, 41, 49, 56, 61, 62, 69, 72, 90, 106, 111, 122, 123, 134, 148, 152, 153, 178, 189, 198, 206, 207, 209, 224, 231, 236, 244, 249, 250, 257, 261, 264, 276, 278, 286, 288, 292